Digital Image Processing Lab (NCSH302) **Session: Monsoon 2026-27**
B. Tech-5th Sem. (CSE-Hons) **Date: 22.09.2026**

Assignment-6 (Image Restoration)

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1. Consider *Blurry_moon.tiff* as the original $8-bit$ grayscale image $f(x,y)$. Generate uniform multiplicative noise $n(x,y)$ in the range $[0.8, 1.2]$ using random $seed$ 0, and obtain the degraded image as $g(x,y) = f(x,y) \times n(x,y)$. Clip all resulting intensity values to the range $[0, 255]$ and convert the degraded image to $8-bit$ format. Since multiplicative noise can be transformed into an additive form using a logarithmic transformation, compute $z(x,y) = log(1 + g(x,y))$, apply a 3×3 arithmetic mean filter to $z(x,y)$, and then recover the restored image using the inverse transformation $\hat{f}(x,y) = e^{(\hat{z}(x,y))} - 1$, where $\hat{z}(x,y)$ is the filtered logarithmic image. Clip the restored image again to $[0, 255]$. Calculate the *Mean Squared Error* (MSE) between the original and degraded images and between the original and restored images. Display the original, degraded, and restored images and report both MSE values rounded to two decimal places.
[**15 Marks**]

Sample Input:



Sample Output:

MSE (original, degraded) = 77.57

MSE (original, restored) = 16.74



2. Consider *Blurry_moon.tiff* as the original $8-bit$ grayscale image $f(x,y)$ and perform two independent restoration experiments. First, starting from the original image, generate salt-and-pepper noise using $P_a = 0.05$ for pepper noise and $P_b = 0.05$ for salt noise with random seed 0, and restore the resulting noisy image using a manually implemented 3×3 median filter. Second, start again from the original image and add sinusoidal periodic noise defined by $n(x,y) = A\,sin[2\pi\,(\frac{u_0\,x}{M} + \frac{v_0 y}{N})]$, using amplitude $A = 40, u_0 = 0, and\ v_0 = 40$, where M and N are the image dimensions. Restore the periodically degraded image in the frequency domain using a fourth-order Butterworth band-reject filter with center radius $D_0 = 40$ and bandwidth $W = 6$. Calculate the PSNR between the original image and the salt-and-pepper noisy image, the median-filtered image, the sinusoidal noisy image, and the Butterworth-restored image. Display the salt-and-pepper noisy image with its median-filtered result and the sinusoidal noisy image with its Butterworth band-reject filtered result and report all four PSNR values rounded to two decimal places.

[15 Marks]

Sample Input:



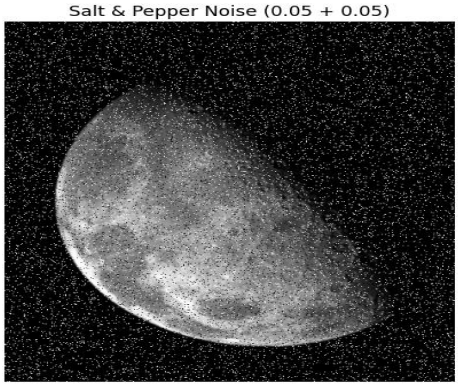
Sample Output:

PSNR - Salt-pepper noisy: 13.61 dB

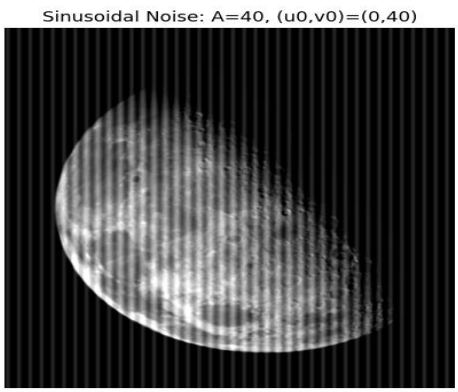
PSNR - Median restored: 39.57 dB

PSNR - Sinusoidal noisy: 20.91 dB

PSNR - Butterworth restored: 38.54 dB





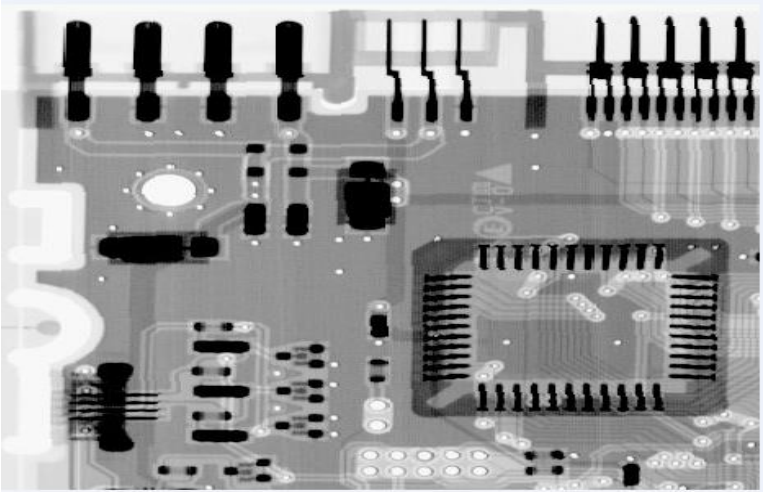




3. Consider *Circuit_Board.tiff* as the original 8-bit grayscale image $f(x, y)$. Generate two independently degraded images using random seed 0: one containing 10% pepper noise, in which randomly selected pixels are replaced by intensity 0, and another containing 10% salt noise, in which randomly selected pixels are replaced by intensity 255. Restore the pepper-noisy image using a 5×5 contraharmonic mean filter with $Q = +1.5$ and restore the salt-noisy image using a 5×5 contraharmonic mean filter with $Q = -1.5$, where the restored value is calculated as $\hat{f}(x, y) = \frac{\sum[g(s,\ t)^{(Q+1)}]}{\sum[g(s,\ t)^{Q}]}$ over the local neighbourhood. Handle zero-valued pixels carefully to avoid division-by-zero or undefined negative powers. Calculate the PSNR between the original image and each noisy image and between the original image and each corresponding restored image. Display the original, pepper-noisy, pepper-restored, salt-noisy, and salt-restored images and report all PSNR values rounded to two decimal places.

[**10 Marks**]

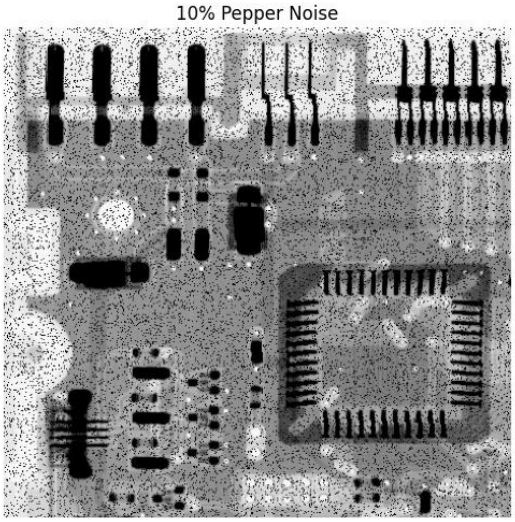
Sample Input:



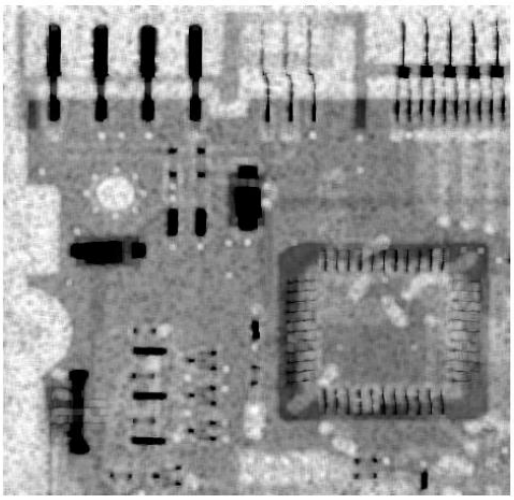
Sample Output:

PSNR - Pepper noisy: 13.75 dB
PSNR - Pepper restored: 17.60 dB
PSNR - Salt noisy: 16.28 dB
PSNR - Salt restored: 16.32 dB

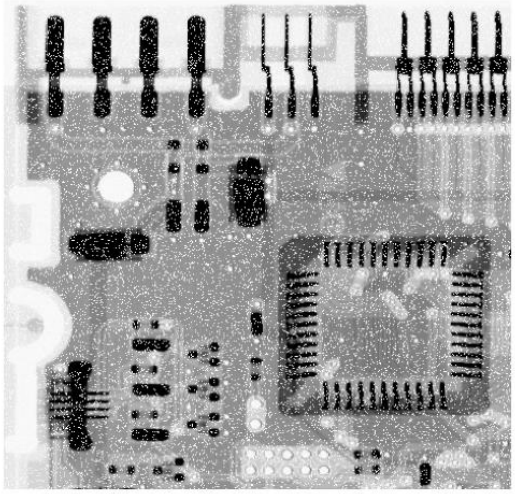
10% Pepper Noise



5x5 Contraharmonic, Q=+1.5



10% Salt Noise



5x5 Contraharmonic, Q=-1.5

